

IMPERIAL



**The London Register of Subsurface
CO₂ Storage
Annual Report 2026**

Overview

We report a summary of the annual amounts of CO₂ stored underground by projects globally over the period 1996–2025. We summarise the methods used to compile this data and provide an overview of the project below. The full Register is maintained as a relational database linking projects, sources, raw observations, and release logs and is available along with the methodology and assessment report through the project website at <https://imperialcollegelondon.github.io/The-London-Register-of-Subsurface-CO2-Storage/>

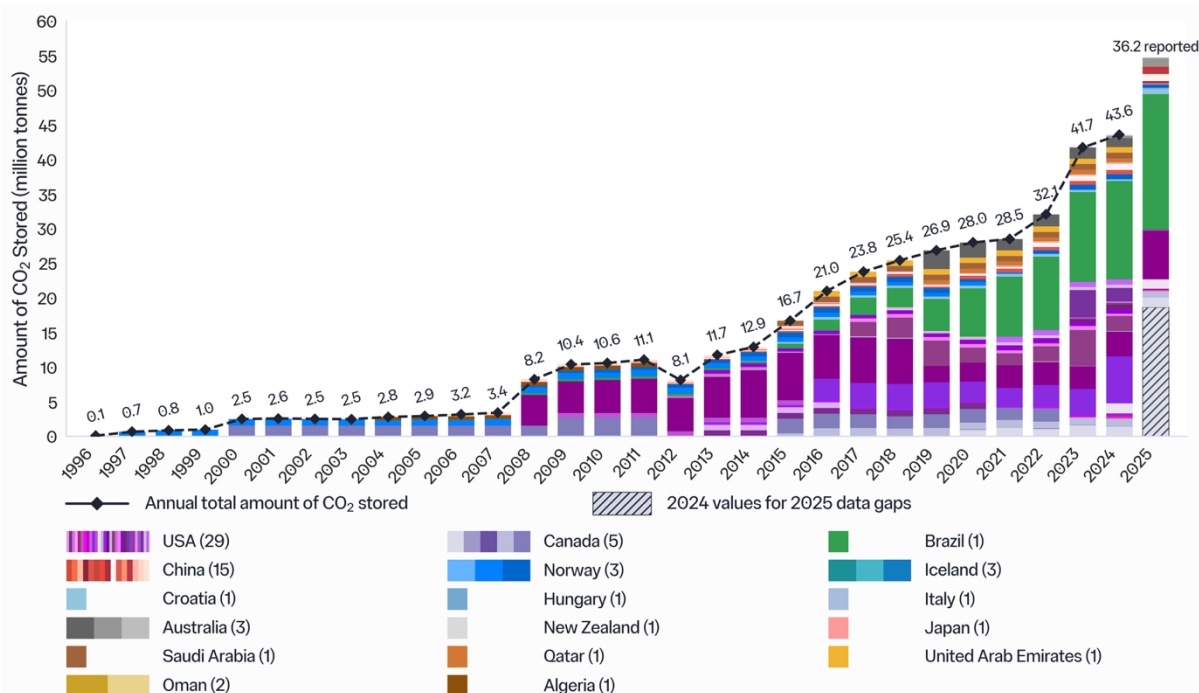


Figure 1. Annual amounts of CO₂ stored with colouring by continent, country, and individual project. See Appendix 1 for the individual project legend. The hashed grey in the bar for 2025 represents the amount of CO₂ stored in 2024 by projects where data has not yet been published

Amounts of CO₂ stored since 1996

From 0.1 million tonnes in 1996, the amount of CO₂ stored globally increased to a high of 43.6 million tonnes in 2024 (Figure 1). Inclusive of 2025, a total of 421.7 million tonnes has been stored, although the 2025 data remain incomplete. Three distinct phases of growth can be observed

- 1. Pioneering Phase (1996–2007, 7 projects):** Annual storage amounts remained below 3.5 million tonnes of CO₂ per year. This phase was driven primarily by early large-scale storage at Sleipner in Norway and Weyburn-Midale in Canada, an early oilfield storage project in China, Zhongyuan, and the In Salah project in Algeria.
- 2. North American and Early International Expansion (2008–2015, 31 projects):** In this phase new projects increased annual reported storage from 8.2 to 16.7 million tonnes of CO₂ per year. This growth was driven by projects in North America and by storage combined with oil recovery. During this time there was the beginning of broader international deployment, including projects in Brazil, China, Australia, Iceland, Croatia, Norway, and Saudi Arabia.
- 3. Global Scaling (2016–2024, 57 projects):** Reported annual storage increased from 21.0 million tonnes in 2016 to 43.6 million tonnes in 2024. This phase reflects both a larger number of operating projects and a broader geographic spread of reported activity. Growth was led by projects in the United States and Brazil, with continued and expanding activity in China, Canada, Australia, Norway, Iceland, Qatar, Saudi Arabia, and the United Arab Emirates.

Amounts of CO₂ stored in 2024

In 2024, the Register records an aggregate reported global storage amount of 43.6 million tonnes of CO₂, an increase of 1.9 million tonnes from the revised 2023 total of 41.7 million tonnes (Figure 1). This comprises records from 49 projects across 13 countries (Figure 2).

Projects in the United States remain the largest contributors to global reported storage, with 23 projects reporting a combined 20.2 million tonnes of CO₂. The net increase from 2023 was 0.6 million tonnes, but this masks larger changes at the project level including a 3.1 million tonne increase at the Denver Unit / Wasson San Andres Field. The 2024 data also includes 1.4 million tonnes from the DGC Beulah Broom Creek project, and increases at Petra Nova West Ranch, Blue Flint CO₂ Storage Project, Barnett RDC Well No. 1, North Burbank Unit, and Shute Creek. These increases were partly offset by lower reported storage at the Hobbs Field and the Seminole San Andres Unit.

Projects in Canada reported 2.5 million tonnes stored in 2024, a small decrease from 2.7 million tonnes reported in 2023. This was mainly due to a lower reported amount from Aquistore and the Clive Sequestration Facility, and offset by a small increase at Quest and by the newly included Glacier Gas Plant.

Storage in Brazil continued to be one of the largest contributors to the global total. The Petrobras Santos Basin Pre-Salt Oil Field reported 14.2 million tonnes in 2024, an increase of 1.2 million tonnes from 2023. This single project accounted for approximately one-third of the global reported storage total in 2024.

Reported storage in China increased from 1.5 million tonnes in 2023 to 1.7 million tonnes in 2024. This growth was driven primarily by storage at the Shengli Oilfield, which increased from 0.85 million tonnes to 0.97 million tonnes, and by the newly included Daqing Aonan Oilfield. Additional reported storage came from projects including the Jilin Oilfield, Karamay Dunhua, the Yanchang Integrated Demonstration, Xinjiang Oilfield, Huabei, and Liaohe.

In Australia 1.7 million tonnes of CO₂ storage was reported. This value is slightly higher than in 2023, with the addition of storage at the Moomba project offsetting a lower reported value at the Gorgon project. In New Zealand, geothermal CO₂ reinjection at Ngawha entered the Register with 0.13 million tonnes.

Projects in the Middle East reported broadly stable values. Projects in Saudi Arabia and the United Arab Emirates each reported 0.8 million tonnes stored, while in Qatar 0.6 million tonnes of storage was reported.

In Europe, storage amounts are dominated by offshore projects in Norway, with 0.7 million tonnes stored at Snøhvit and 0.05 million tonnes stored at Sleipner. Reporting also continued in Croatia at 0.2 million tonnes, and Italy entered the Register with 0.03 million tonnes at the ENI Ravenna project. However, this is joined by reporting from the Orca and Mammoth projects in Iceland, which combined reported 0.0011 million tonnes of stored CO₂.

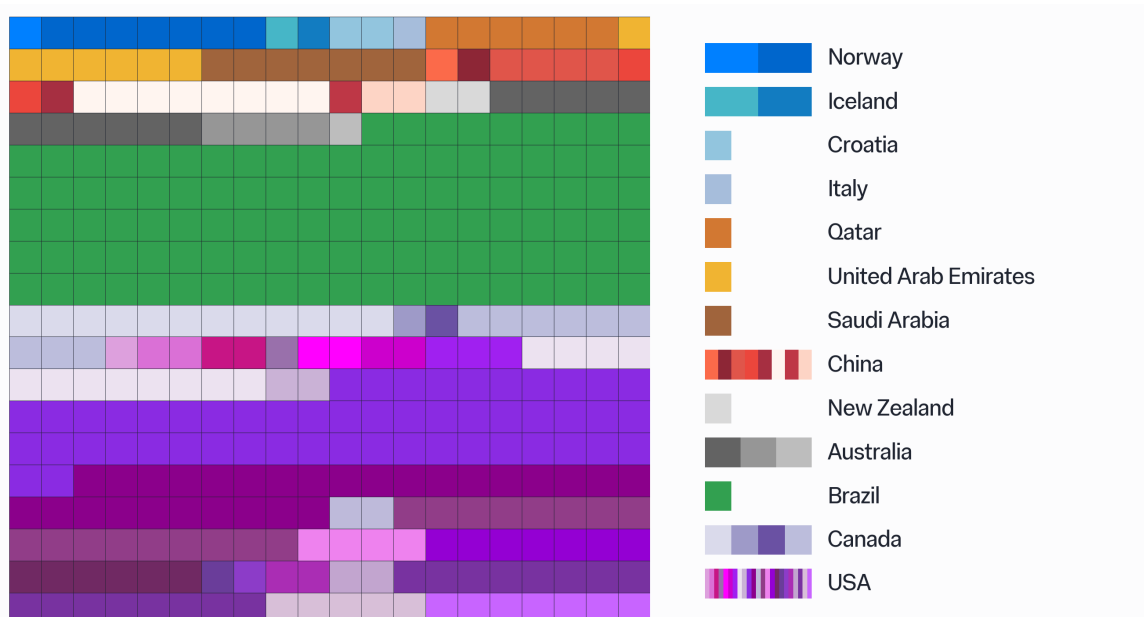


Figure 2. A waffle Plot of 2024 CO₂ Storage by country and project. The individual project legend is provided in Appendix 1.

Incomplete Reporting on Amounts of CO₂ Stored in 2025

At the time of publication, 29 projects across 12 countries had reported 2025 storage data, with an aggregate amount of 36.2 million tonnes of CO₂. These data are currently incomplete and should not be interpreted as a full-year global total.

The largest reported contribution for 2025 is from Brazil. The Petrobras Santos Basin Pre-Salt Oil Field reported 19.6 million tonnes of CO₂ stored in 2025, an increase from 14.2 million tonnes in 2024.

In China, 2.6 million tonnes of CO₂ stored has been reported for 2025 by seven projects. The largest contributions are from Xinjiang Oilfield, Shengli Oilfield, and Liaohe, with additional reported storage from Tuha, Enping, Daqing Aonan, and Huabei.

Reporting for 2025 from the United States currently includes 8.9 million tonnes of CO₂ stored across 6 projects, including the ExxonMobil Aggregate Projects (7.0 million tonnes reported as captured for storage at company level, pending reconciliation with project-level records — see Back Revisions), DGC Beulah Broom Creek (1.4), Blue Flint CO₂ Storage Project (0.2), Red Trail Energy (0.2), Barnett RDC Well No. 1 (0.1), and Core Energy Otsego County EOR Operations (0.06). Several U.S. projects with 2024 data do not yet have corresponding 2025 records in the Register, including large contributors such as the Denver Unit / Wasson San Andres Field, Hobbs Field, Seminole San Andres Unit, North Burbank Unit, Petra Nova West Ranch, Shute Creek, and West Seminole San Andres Unit.

In Canada, the Quest project has reported 0.89 million tonnes stored and Aquistore 0.04 million tonnes, while a 1.38 million tonne record for the Clive Sequestration Facility has also been added.

In Australia, the Moomba Carbon Capture and Storage project has reported 1.2 million tonnes stored in 2025. Other Australian projects with 2024 records, including Gorgon and Otway, do not yet have corresponding 2025 records in the Register. In New Zealand, the Ngawha Geothermal Reinjection project reported 0.13 million tonnes in 2025.

In Europe, reported storage for 2025 includes 0.5 million tonnes in Norway, where the Northern Lights project began injection in August 2025; 0.7 million tonnes in Croatia; 0.16 million tonnes at the Szank Field in Hungary; and 0.03 million tonnes at the ENI Ravenna project in Italy. The Orca and Mammoth projects in Iceland have reported 0.0014 million tonnes of stored CO₂. In Oman, the Hajar and Moringa mineralisation projects entered the Register with a combined 0.03 million tonnes.

The Role of Enhanced Oil Recovery

A large share of the CO₂ reported in the Register is associated with enhanced oil recovery (EOR) (Figure 3). The 35 project records classified as Geological Storage with Oil Recovery, comprising 34 individual projects and one company-level aggregate, account for 347.2 million tonnes, or 82.3% of the cumulative Register total of 421.7 million tonnes, including the incomplete data reported for 2025. In 2024, EOR records accounted for 36.2 million tonnes, or 83.0% of the global reported total of 43.6 million tonnes, while dedicated geological storage increased from 5.5 million tonnes in 2023 to 7.4 million tonnes in 2024. These figures describe the reported historical mix and do not determine the future balance between storage routes.

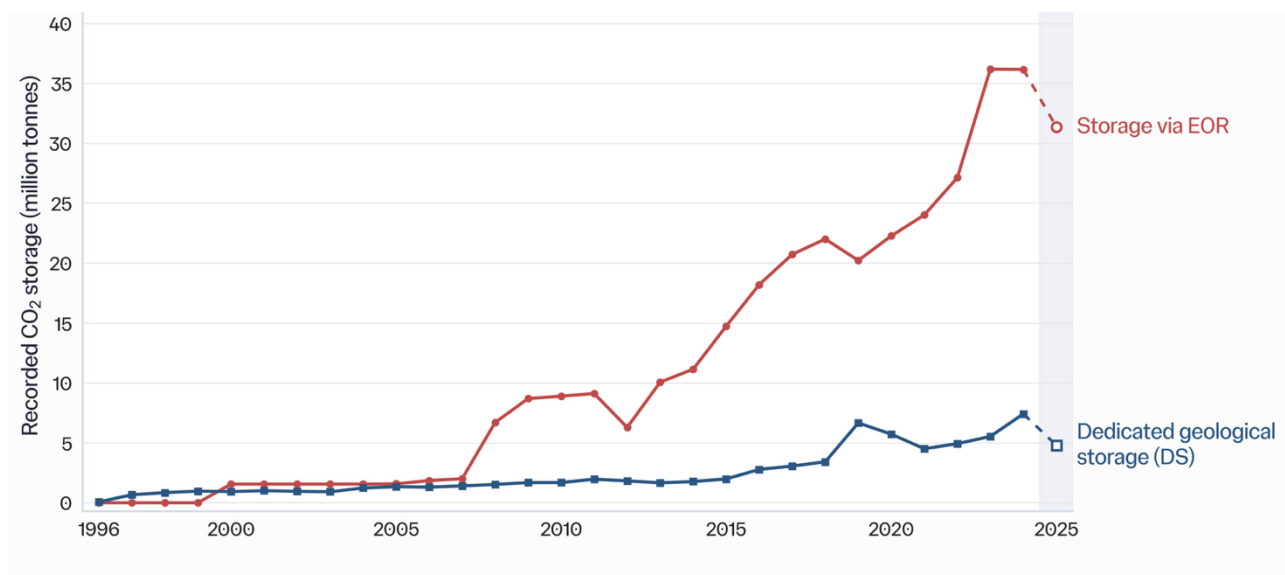


Figure 3. Annual reported CO₂ storage via enhanced oil recovery (EOR) and dedicated geological storage (DS), 1996–2025. Reporting for 2025 remains incomplete and is not directly comparable with complete reporting years.

U.S. EOR projects reporting under the Environmental Protection Agency’s Greenhouse Gas Reporting Program Subpart RR quantify net storage through mass-balance accounting under EPA-approved monitoring, reporting and verification (MRV) plans. The framework accounts for CO₂ injected, produced and emitted, including surface leakage, to avoid counting recycled CO₂ repeatedly as additional storage. Records in this category account for 70.1 million tonnes, or 20.2% of cumulative EOR, including 13.53 million tonnes across 11 projects in 2024.

Other records with a documented net-storage basis account for 56.2 million tonnes, or 16.2% of cumulative EOR, and 1.81 million tonnes in 2024. Supporting evidence includes the subtraction of recirculated CO₂ from gross injection in regulator reporting for the Clive Sequestration Facility, and Weyburn’s separate reporting of injected and stored quantities. Accounting and monitoring arrangements differ between projects.

For the remaining records, the accounting basis is less explicit. Excluding Subpart RR records, quantities described as CO₂ injected without an explicit adjustment for produced and reinjected

CO₂ account for 112.5 million tonnes, or 32.4% of cumulative EOR. In 2024, this category comprised 15.45 million tonnes, or 42.7% of annual EOR. The Petrobras Santos Basin Pre-Salt Oil Field contributed 14.18 million tonnes, representing 91.8% of this category. Petrobras treats its reported reinjected quantity as stored CO₂, but the cited material does not explicitly explain how produced and recycled CO₂ are accounted for.

EOR records whose sources do not state the accounting basis account for 100.9 million tonnes cumulatively and 4.58 million tonnes in 2024, or 12.7% of annual EOR. A further 7.5 million tonnes cumulatively and 0.80 million tonnes in 2024 are associated with CO₂ described as received or delivered for injection, without establishing a net-storage quantity.

ExxonMobil's company-level aggregate is classified as EOR in this release, with no accounting basis stated. After adjustment for overlap with separately reported projects, it contributes 88.4 million tonnes cumulatively and 3.57 million tonnes in 2024. The 2024 amount represents 78.0% of the EOR category with no stated accounting basis and 8.2% of the global reported total. Its underlying disclosure describes CO₂ captured for storage, rather than an explicitly established net geological-storage quantity.

Back Revisions and Changes in this Report

This year's publication includes several updates and changes to the processed data from the previous annual report. For 1996–2008, the revisions mainly result from differences in the rounding of values for the Sleipner project and the addition of the In Salah project in Algeria (2004–2011). For 2009–2015, the principal changes are the addition of the Cranfield, Bell Creek, Citronelle, Air Products Port Arthur and Core Energy Otsego projects, partly offset from 2013 by a new interpretation of data from the North Burbank Unit to account for recycled CO₂. For 2016–2024, the revisions primarily reflect the continued adjustment at North Burbank, a correction in the interpretation of the raw values at Aquistore as cumulative, a correction of some double counting between ExxonMobil's company-level and project-level reporting, the reallocation of the Jilin Oilfield series to the operator's cumulative disclosures, the adoption of regulator-reported values for the Alberta Carbon Trunk Line, and the addition of more complete 2024 data. The Sleipner value for 2021 has also been corrected from 0.322 to 0.260 million tonnes, following the revised Norwegian National Inventory Document. A large amount of previously unavailable data for U.S. projects reporting under EPA Greenhouse Gas Reporting Program Subpart RR in 2024 became available through a third party Freedom of Information Act request. Relative to the figures cited for the 2025 Annual Report, these revisions reduce cumulative reported storage through 2023 from 351.4 to 342.0 million tonnes, a decrease of 2.7%. The reported amount for 2023 decreases from 45.2 to 41.7 million tonnes, a reduction of 7.7%.

There are several structural changes to the Register. The data collection methodology has been formalised and is published in a methodology report. The architecture of the database has also been changed from a spreadsheet to a relational database structure. The core database links project records, source records, raw observations, and version logs, allowing each reported value to be traced to a project, source, source type, and release record. This structure supports the updated processing of observations and makes back revisions more easily traceable.

Methods

The London Register of Subsurface CO₂ Storage compiles reported amounts of CO₂ stored geologically. We summarise the methods used to compile this data here. The formal methodology can be downloaded from the project website <https://imperialcollegelondon.github.io/The-London-Register-of-Subsurface-CO2-Storage/>.

- Data sources: The data in this report is compiled from publicly reported sources, including project-level disclosures, government databases, industry reports, environmental

assessments, and data directly from project operators by response to a survey. We have archived copies of all primary sources of data.

- **Data processing:** The Register is maintained as a relational database linking projects, sources, raw observations, and release logs. For each project-year, the Register retains all raw observations identified from the source material, including values from multiple sources and both annual and cumulative records where these are reported. For summarizing calculations published in this report and on our website where we provide single values for projects in a year, we assign a priority record in the relational database, generally prioritising project-level primary sources such as operator survey responses or government reporting over secondary reporting. Where annual and cumulative records overlap, a project-year is counted only once to avoid double counting. For projects where only cumulative amounts of stored CO₂ are reported over several years of activity, we distribute the cumulative value across the relevant years following the documented processing method. Full records and processing details are available in the Register database and methodology report.
- **Definitions of storage:** We accept definitions of storage as reported by projects, and these are usually governed by adherence to independent standards, regulatory reporting programmes, and verification schemes. The standard under which each value is reported is recorded against the individual observation in the Register database. Links to these frameworks are provided in Appendix 2.
- **Assurance and Quality:** An independent third party assessment of the data quality and management practice of the Register has been carried out by DNV. The detailed outcome and recommendations can be found in a separate report available on the website.

About the London Register of Subsurface CO₂ Storage

This is a centralised register of annual rates of CO₂ stored underground from operational projects worldwide. This project is supported by the Royal Academy of Engineering under the Research Chairs and Senior Research Fellowships programme, and a consortium of stakeholders from academia, industry, and government. The Register compiles data obtained from operators and from publicly available information, providing a standardised database of the annual amounts of stored CO₂.

This report should be cited as:

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Management

Prof. Samuel Krevor – Director, Professor of Subsurface Carbon Storage, Imperial College London

Dr. Xiaowei Gao – Executive Director, Postdoctoral Researcher, Imperial College London

Dr. Anjanette Harry – Programme Manager, Imperial College London

Maksim Elizarev – Quality control for data input, Postdoctoral Research Assistant, Imperial College London

Consortium Members (updated in Sep 2026)

Dr Juan Alcalde – Tenured Scientist, GEO3BCN-CSIC

Prof. Sally Benson – Professor of Energy Science & Engineering, Stanford University

Dr Andrew Cavanagh – Scottish Carbon Capture & Storage, Edinburgh Climate Change Institute, University of Edinburgh

Dr Nicola Clarke – Senior Geologist, IEAGHG

Chris Consoli – Principal and Head of Quantitative Analysis and Research, Global CCS Institute

Jarad Daniels – CEO, Global CCS Institute

Nihal Darraj – Research Associate at Centre for Environmental Policy, Imperial College London

Tim Dixon – Director and General Manager, IEAGHG

Prof. Sarah Gasda – Research Director, NORCE

Peter Golding – Program Manager, Oil and Gas Climate Initiative

Sarah Hardman – Partnerships Manager & Consortia Lead, Global CCS Institute

Dr Chris Holdsworth – Technical Specialist in Measurement, Reporting, and Verification, Carbfix

Dr Samuel Jackson – Senior Research Scientist, CSIRO

Prof. Ruben Juanes – Professor of Civil and Environmental Engineering, MIT

Anhar Karimjee – Head of Strategic Planning and Business Integration, Global CCS Institute

Douglas Lau – Associate, Low Carbon Solutions, Oil and Gas Climate Initiative

John MacArthur – Royal Academy of Engineering/Highland Earth

Dr Shehadeh Masalmeh – EOR and SCAL Expert, ADNOC

Laura Mislan – Senior Development Lead, Enhance Energy

Dr Rachael Moore – Managing Director, CarbStrat

Zeinab Movahedzadeh – Project Manager, Petroleum Technology Research Centre

Erik Nickel – Chief Operating Officer and Interim CEO, Petroleum Technology Research Centre Inc

Candice Paton – Vice President, Corporate Affairs, Enhance Energy

Prof. Philip Ringrose – Professor in Energy Transition Geoscience, Norwegian University of Science and Technology

Dr Jennifer Roberts – Senior Lecturer, Department of Civil & Environmental Engineering, University of Strathclyde; Deputy Director, UK Carbon Capture and Storage Research Centre (UKCCSRC)

Traci Rodosta – Strategy and Advocacy Advisor Americas, Global CCS Institute

Dr Andrew Ross – Group Leader, CSIRO

Justine Roure – Vice President Strategy & Low Carbon Solutions, Oil and Gas Climate Initiative

Norm Sacuta – Director of Communications, Petroleum Technology Research Centre

Alberto Sanchez – Policy Manager, Carbon Capture and Storage Association

Hailey Solie – Senior Specialist, Contracts and Finance, Petroleum Technology Research Centre

Dr James White – Research Geophysicist, British Geological Survey

Prof. Ruina Xu – Professor in Thermal Engineering, Department of Energy and Power Engineering, Tsinghua University

Appendix 1: Project Legend for Figure 1 and Figure 2

North America

Canada:

-  Alberta Carbon Trunk Line (ACTL)
-  Aquistore
-  Glacier Gas Plant
-  Quest
-  Weyburn-Midale

USA:

-  30-30 Gas Plant
-  Air Products Port Arthur
-  Barnett RDC Well No. 1
-  Bell Creek Field Project
-  Blue Flint CO₂ Storage Project
-  Booker Field Area
-  Campo Viejo Gas Processing Plant
-  Camrick Unit
-  Citronelle Project (Plant Barry)
-  Core Energy Otsego County EOR
-  Cranfield
-  DGC Beulah Broom Creek
-  Dark Horse Treating Plant
-  Denver Unit/Wasson San Andres
-  ExxonMobil Aggregate Projects
-  Farnsworth Unit CO₂ Flood
-  Hobbs Field
-  Illinois Basin Decatur Project
-  Illinois Industrial CCS Project
-  North Burbank Unit
-  Petra Nova West Ranch
-  Rangely CO₂ Flood
-  Red Hills Gas Processing Plant
-  Red Trail Energy, LLC
-  SPG CO₂ Bowie Facility
-  Seminole East Field (SEF)
-  Seminole San Andres Unit
-  Shute Creek
-  West Seminole San Andres Unit

South America

Brazil:

-  Petrobras Santos Basin Pre-Salt Oil Field

Europe

Croatia:

-  CO₂ EOR Project Croatia

Hungary:

-  MOL Szank Field CO₂-EOR

Iceland:

-  Carbfix Storage
-  Mammoth
-  Orca

Italy:

-  ENI Ravenna CCS Phase 1

Norway:

-  Northern Lights
-  Sleipner
-  Snøhvit

Africa

Algeria:

-  In Salah

Asia

China:

-  Caoshe Oilfield
-  Daqing Aonan Oilfield
-  Enping 15-1 Offshore CO₂ Storage
-  Huabei Oilfield
-  Jilin Oilfield
-  Jingbian Oilfield
-  Karamay Dunhua
-  Liaohe Oilfield
-  Shengli Oilfield
-  Shenhua Ordos
-  Tuha Oilfield
-  Xinjiang Oilfield
-  Xinjiang Ordos Basin
-  Yanchang Integrated Demonstration
-  Zhongyuan Oilfield

Japan:

-  Tomakomai CCS Demo Project

Oman:

-  44.01 Project Hajar
-  Sirona Project Moringa

Qatar:

-  Qatar LNG

Saudi Arabia:

-  Uthmaniyah CO₂-EOR Demonstration

United Arab Emirates:

-  ADNOC Al Reyadah

Oceania

Australia:

-  Chevron Gorgon
-  Moomba CCS
-  Otway International Test Centre

New Zealand:

-  Ngawha Geothermal Reinjection

6 Continents | 17 Countries | 70 Projects

Appendix 2: Reporting Standards Used by CO₂ Storage Projects in the Register

A variety of reporting standards are used in the raw data that was compiled (Table A2.1). Reporting standards comprise three regulatory, compliance, and exchange reporting frameworks for statutory or listing-rule disclosures; four voluntary corporate sustainability and climate-disclosure frameworks for company-level ESG, sustainability and investor-facing reporting; one national greenhouse gas inventory methodology for UNFCCC inventory reporting; and two further references covering energy management systems and carbon-removal certification. Additionally, just under one third of the cumulative stored amounts compiled did not publish a standard by which they reported.

Among the reporting standards, the EPA Greenhouse Gas Reporting Program Subpart RR is the only framework that applies specifically to site-level CO₂ storage and requires an EPA-approved monitoring, reporting and verification plan. The other references generally operate at company, issuer, national-inventory, energy-management or certification level. Accordingly, the reference cited against each observation records how the stored-CO₂ value was reported and defined, rather than providing independent assurance of that value. Each reference listed below is cited by at least one observation in the Register database, and Table A2.1 and Figure A2.1 summarise the projects, countries and observations citing each reference.

Table A2.1. Register projects and observations citing each reporting reference

Reporting standard	Type	Projects	Observations	Countries	Cumulative Stored [Mt]
EPA GHGRP Subpart RR (U.S.)	Regulatory reporting	23	86	USA	83.1
GRI Standards	Voluntary disclosure	5	39	Brazil, Canada, China, Qatar	64.7
Ipieca guidance (CCE indicators)	Voluntary disclosure	3	37	Canada, China, USA	97.6
IPCC 2006 Guidelines (via UNFCCC national inventories)	National-inventory methodology	5	61	Croatia, Iceland, Japan, Norway	27.8
HKEX ESG Reporting Code	Exchange listing-rule disclosure	2	12	China	5.6
ISO 50001	Management-system standard	2	11	Saudi Arabia, United Arab Emirates	7.2
Puro Standard (Geologically Stored Carbon)	Carbon-removal certification	2	7	Iceland	<0.1
SASB Standards (EM-EP)	Voluntary disclosure	2	6	Canada	5.8
NGER (Australia)	Regulatory reporting	1	9	Australia	11.2
TCFD recommendations	Voluntary disclosure	1	3	Canada	0.1
No reporting standard recorded	—	46	251	Algeria, Australia, Brazil, Canada, China, Croatia, Hungary, Italy, New Zealand, Norway, Oman, Saudi Arabia, USA	133.4

Register total	70	491	17 countries	421.7
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Notes: Stored CO₂ is the cumulative mass in the published series shown in the report figures, including values reported to date for 2025, associated with the observations citing each reference. An observation citing several references is counted once under each, so the rows are not additive.

Further context and links to the reporting standards are provided here.

Regulatory, compliance and exchange-reporting frameworks

- **EPA GHGRP – EPA Greenhouse Gas Reporting Program, Subpart RR – Geologic Sequestration of Carbon Dioxide (U.S. Environmental Protection Agency)** [<https://www.epa.gov/ghgreporting/subpart-rr-geologic-sequestration-carbon-dioxide>] — Mandatory U.S. federal reporting rule (40 CFR Part 98) for facilities that inject CO₂ underground for geologic sequestration, including requirements for EPA-approved monitoring, reporting and verification (MRV) plans and annual reporting of the mass of CO₂ received, injected, produced, emitted or sequestered.
- **NGER – National Greenhouse and Energy Reporting scheme (Australia, Clean Energy Regulator)** [<https://cer.gov.au/schemes/national-greenhouse-and-energy-reporting-scheme>] — Australia’s mandatory national reporting framework, established under the National Greenhouse and Energy Reporting Act 2007, for corporate greenhouse gas emissions, energy production and energy consumption by entities that meet statutory reporting thresholds.
- **HKEX ESG Reporting Code, Appendix C2 to the Main Board Listing Rules (Hong Kong Stock Exchange; formerly the ESG Reporting Guide, Appendix 27)** [<https://en-rules.hkex.com.hk/rulebook/appendix-c2-environmental-social-and-governance-reporting-code-0>] — ESG disclosure requirements for Hong Kong-listed issuers, combining mandatory disclosures with comply-or-explain environmental key performance indicators covering emissions, resource use, waste and environmental management. This is issuer-level ESG disclosure rather than a CO₂ storage-specific MRV standard.

Voluntary corporate sustainability and climate-disclosure frameworks

- **GRI Standards, including GRI 305: Emissions 2016 and GRI 11: Oil and Gas Sector 2021** [<https://www.globalreporting.org/how-to-use-the-gri-standards/gri-standards-english-language/>] — Voluntary sustainability impact-reporting standards used by organisations to disclose economic, environmental and social impacts, including emissions-related disclosures and oil-and-gas sector-specific topics; each GRI standard carries its own version year.
- **Ipieca Sustainability Reporting Guidance for the Oil and Gas Industry, including Climate Change and Energy (CCE) indicators such as CCE-3** [<https://www.ipieca.org/resources/sustainability-reporting-guidance>] — Voluntary oil-and-gas sector sustainability reporting guidance published by Ipieca jointly with the American Petroleum Institute and the International Association of Oil & Gas Producers, providing

indicators — including the climate change and energy (CCE) series — used to structure corporate disclosures on emissions, energy use and climate-related performance.

- **SASB Standards, Oil & Gas – Exploration & Production (EM-EP)** [<https://www.ifrs.org/issued-standards/sasb-standards/>] — Industry-specific, investor-focused sustainability disclosure standards covering financially material sustainability topics and metrics for oil and gas exploration and production companies; now maintained by the International Sustainability Standards Board under the IFRS Foundation.
- **TCFD – Recommendations of the Task Force on Climate-related Financial Disclosures** [<https://www.fsb-tcfd.org/recommendations/>] — Voluntary climate-related financial disclosure framework structured around governance, strategy, risk management, and metrics and targets, including disclosure of greenhouse gas emissions and climate-related risks and opportunities; the Task Force was disbanded in October 2023 and monitoring of climate-related disclosures passed to the IFRS Foundation.

National greenhouse gas inventory methodologies

- **IPCC 2006 Guidelines for National Greenhouse Gas Inventories (Intergovernmental Panel on Climate Change)** [<https://www.ipcc-nggip.iges.or.jp/public/2006gl/>] — Methodological guidance for compiling national greenhouse gas inventories, forming the basis of national inventory reporting under the UNFCCC; Volume 2, Chapter 5 covers CO₂ transport, injection and geological storage. In the Register it underpins the National Inventory Documents used for Iceland, Japan, Norway and Croatia.

Energy management and carbon-removal certification references

- **ISO 50001 – Energy Management Systems (International Organization for Standardization)** [<https://www.iso.org/iso-50001-energy-management.html>] — International energy management system standard that provides a framework for organisations to improve energy performance, establish energy policies and targets, use energy data, and support continual improvement in energy management; referenced in the Register through a Clean Energy Ministerial case study.
- **Puro Standard – Geologically Stored Carbon methodology and General Rules (Puro.earth)** [<https://puro.earth/methodologies/geologically-stored-carbon/>] — Carbon-removal crediting methodology and certification framework for geologically stored carbon, including independent verification and issuance of CO₂ Removal Certificates through the Puro Registry.

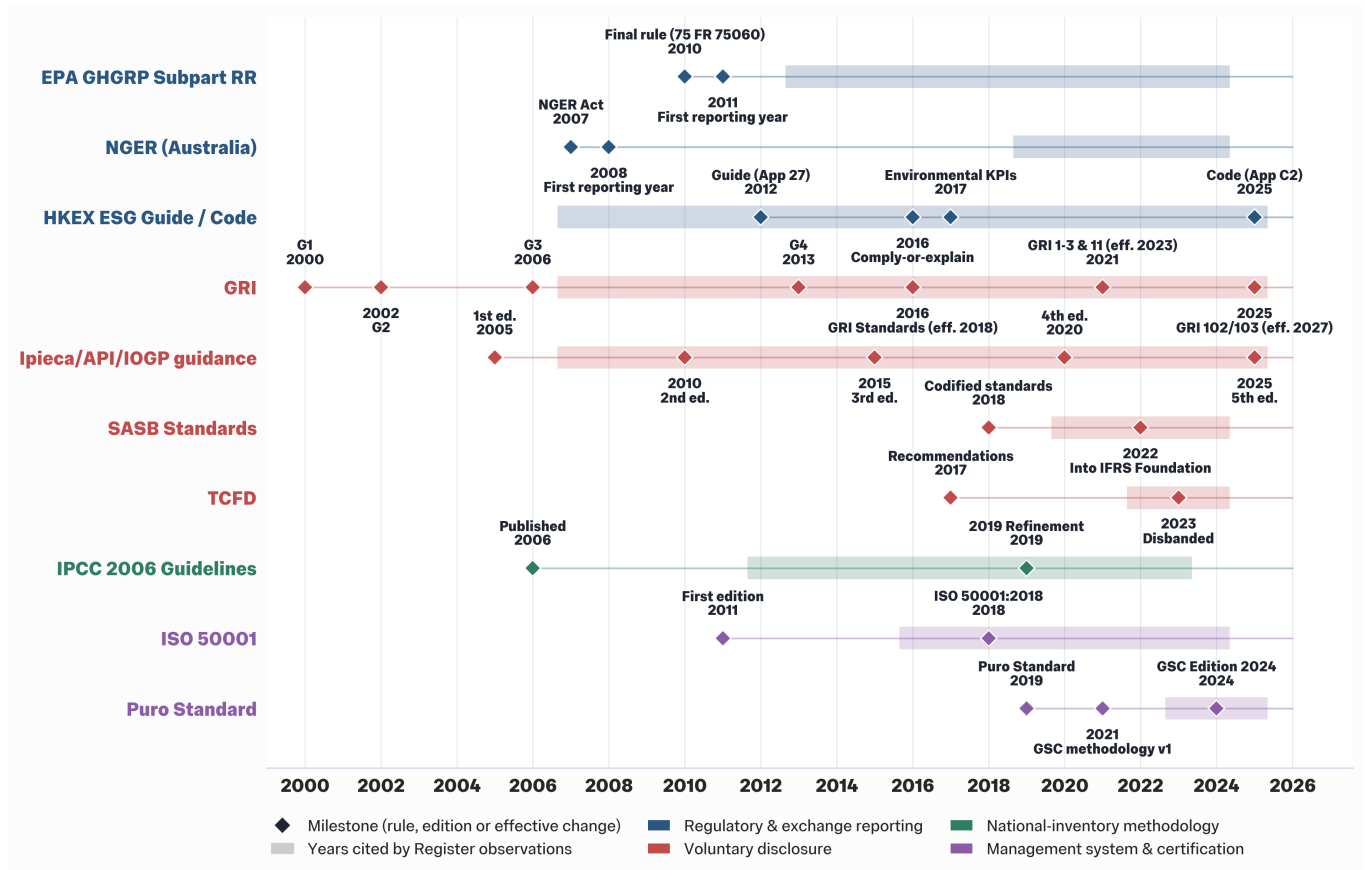


Figure A2.1. Timeline of the ten reporting references cited in the Register. Diamonds mark publication, edition and effective dates as stated by the issuing organisations, and shaded bands show the years cited by Register observations. Details are given in Table A2.1.